

Serhii Kuzmin

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Education

The University of Scranton, Scranton, PA

Bachelor of Science in Computer Science and Mathematical Sciences (Double Major)

May 2026

GPA: 3.86/4.00

- Dean's List: 8/8 semesters
- Honors Society: Upsilon Pi Epsilon, Pi Mu Epsilon

Technical Skills

Languages: Python, Java, Kotlin, C, TypeScript, SQL

Backend Framework & Libraries: FastAPI, Node.js, Bun.js, Python CFFI, REST APIs

Systems & Infrastructure: Git, Docker, Linux, AWS, GitHub/GitLab, GitHub Actions

Databases: PostgreSQL, SQLite, MongoDB

Projects

Surface Evolver Desktop Application

January 2026 – July 2026

- Achieved ~90% backwards compatibility for a legacy mathematical engine by developing a cross-platform Tauri native desktop application wrapping 50 core engine functions.
- Designed a custom worker-lifecycle and mutex-serialization layer using a Bun-compiled FFI sidecar process per session, bypassing a critical single-initialization restriction in the underlying C library.
- Developed the RPC dispatch layer handling session, file, and persistence operations, including a background auto-snapshot/restore system that checkpoints engine state after every mutating command.
- Built and maintained a release pipeline: CMake for building a headless library, a GitHub Action CI Matrix for producing platform installers for Mac, Windows and Linux, and the three-part test suite CTest for the C API, cargo check for the Rust backend, tsc for the frontend

Recipe Diary Android Application

April 2026 – May 2026

- Designed and implemented the local data persistence layer from scratch — SQLite database schema and Room ORM data access layer — supporting recipe creation, management, and offline-first storage.
- Integrated remote data synchronization with an external recipe and meal data provider via a Retrofit-based REST API client, handling network communication between the app and the third-party service.
- Built the application using MVVM architecture in Kotlin with Jetpack Compose, with test coverage across unit, integration, and end-to-end layers validating data integrity and API integration behavior

Full-Stack Group Chat Application

September 2024 – December 2024

- Designed and implemented a full-stack group chat application in C using a client-server architecture.
- Designed and implemented custom communication protocol to speed up the communication by 25% via WebSockets.
- Deployed the server on an AWS-hosted FreeBSD instance with Docker for cross-platform compatibility.

Research Experience

The University of Scranton, Scranton, PA

September 2025 – May 2026

Independent Quantum Computing Student Researcher

- Created hybrid quantum simulations using Python with Qiskit to run and verify experiments.
- Investigated the numerical stability of 1 and 2-qubit quantum Z gates that implement the Variational Quantum Eigensolver algorithm by modifying the readout error in increments of 5%.

Professional Experience

The University of Scranton, Scranton, PA

September 2025 – May 2026

Work Study Assistant

- Promoted Global Education programs through workshops and student engagement.
- Managed budget, spending, and travel requests for the International Students Club.

The University of Scranton, Scranton, PA

October 2023 – December 2025

Peer Tutor

- Provided personalized tutoring in mathematics and computer science through one-on-one instruction tailored to individual learning needs.
- Improved students' average academic performance by approximately 30% through customized study plans and targeted skill development.
- Facilitated drop-in tutoring sessions for both individual students and small groups, addressing diverse academic challenges in real-time.